Problem Set 8

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: ISL 3, 4, 6, ILA 4.3, UDL 5

Reminders:

1. Please come to see me (or Maeve) during my office hours if you are struggling with the quizzes or assignments
2. I am happy to make myself available at other times of the current hours do not work for you
3. Please refrain from using AI to answer any of the problems on any of the assignment
4. You need to be able to explain and defend all solutions that you submit
5. This entire assignment has an extra 30%. To get at least 100%, you can pick one of the following options:
   (a) Skip Problems 7 and 8 and solve Problems 9 and 10
   (b) Skip Problems 9 and 10 and solve Problems 7
   (c) Solving Problem 8 alone gets you 130% (you still need to be able to solve the other problems on quizzes)

Problems

1. Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$.
   (a) Prove that $\boldsymbol{A}^\top \boldsymbol{A}$ is always positive semidefinite.
   (b) Prove (sufficient to outline the argument) that $\boldsymbol{A}^\top \boldsymbol{A}$ is positive definite if and only if the columns of $\boldsymbol{A}$ are linearly independent.
2. **[code]** Consider the following five data points (t_i, y_i):

t_i	1	2	3	4	5
y_i	2.1	3.9	6.2	7.8	10.1

   You will fit the simple linear regression model $y \approx x_1 + x_2 t$, where x_1 is the intercept and x_2 is the slope. Solve this problem *from scratch* using only basic matrix operations (such as matrix multiplication and solving a linear system); do **not** use a linear regression or curve-fitting library (such as `GLM.jl`, `scikit-learn`, `numpy.polyfit`, `numpy.linalg.lstsq`, or R's `lm`).
   (a) Construct the matrix $\boldsymbol{A} \in \mathbb{R}^{5 \times 2}$ and the vector $\boldsymbol{b} \in \mathbb{R}^5$ so that fitting the model is equivalent to minimizing $\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|^2$ over $\boldsymbol{x} = (x_1, x_2)^\top$. Write out $\boldsymbol{A}$ and $\boldsymbol{b}$ explicitly.
   (b) Compute the least-squares solution $\boldsymbol{x}^\star$ by solving the normal equations $\boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}^\top \boldsymbol{b}$. Report the intercept, the slope, and the residual sum of squares $\|\boldsymbol{A}\boldsymbol{x}^\star - \boldsymbol{b}\|^2$.
   (c) Plot the five data points together with the fitted line $y = x_1^\star + x_2^\star t$.

(d) Now consider a data set with three features t_1, t_2, t_3 and only three data points:

i	t_{i1}	t_{i2}	t_{i3}	y_i
1	1	2	0	3
2	0	1	3	5
3	2	1	1	4

Repeat parts (a) and (b) for the linear regression model with an intercept, $y \approx x_1 + x_2 t_1 + x_3 t_2 + x_4 t_3$ (there is no need for a plot). What do you observe when you try to solve the normal equations? Explain why this happens.

(e) For the data set in part (d), can you find a vector $\boldsymbol{x}$ that minimizes $\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|^2$? Is there more than one? Can you find them?

3. Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$, let $\boldsymbol{x}^\star$ be a solution of the normal equations $\boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}^\top \boldsymbol{b}$, and let $\boldsymbol{r} = \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}^\star$ be the vector of residuals.

(a) Show that $\boldsymbol{A}^\top \boldsymbol{r} = \boldsymbol{0}$, that is, the residual vector is orthogonal to every column of $\boldsymbol{A}$.

(b) Suppose that the first column of $\boldsymbol{A}$ is all ones, as in a model with an intercept. Show that the residuals sum to zero: $\sum_{i=1}^m r_i = 0$.

(c) Use part (b) to show that the fitted line $y = x_1^\star + x_2^\star t$ in Problem 2 passes through the point of averages $(\bar{t}, \bar{y})$, where $\bar{t} = \frac{1}{m}\sum_{i=1}^m t_i$ and $\bar{y} = \frac{1}{m}\sum_{i=1}^m y_i$. Verify this numerically using the intercept and slope you computed in Problem 2.

4. Let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ be a symmetric positive semidefinite matrix, that is, $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} \geq 0$ for all $\boldsymbol{x} \in \mathbb{R}^n$. Show that for any $\epsilon > 0$, the matrix $\boldsymbol{A} + \epsilon \boldsymbol{I}$ is positive definite, that is, $\boldsymbol{x}^\top (\boldsymbol{A} + \epsilon \boldsymbol{I})\boldsymbol{x} > 0$ for all $\boldsymbol{x} \neq \boldsymbol{0}$. Then show that $\boldsymbol{A} + \epsilon \boldsymbol{I}$ is invertible. *Hint*: Can you use the eigendecomposition $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top$ to construct the inverse of $\boldsymbol{A} + \epsilon \boldsymbol{I}$?

5. Recall that linear regression (least squares) computes the coefficients $\boldsymbol{x} \in \mathbb{R}^n$ by minimizing $\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|^2$, where $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ is the matrix of features and $\boldsymbol{b} \in \mathbb{R}^m$ is the vector of responses. Ridge regression adds a penalty on the size of the coefficients and instead minimizes the function $f \colon \mathbb{R}^n \to \mathbb{R}$ defined as

$$f(\boldsymbol{x}) = \|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|^2 + \lambda \|\boldsymbol{x}\|^2,$$

for some regularization parameter $\lambda > 0$.

(a) Derive the gradient and the Hessian of f, and show that f is convex.

(b) Use the first-order optimality conditions to show that the ridge regression solution is $\boldsymbol{x}^\star = (\boldsymbol{A}^\top \boldsymbol{A} + \lambda \boldsymbol{I})^{-1} \boldsymbol{A}^\top \boldsymbol{b}$.

(c) Show that $\boldsymbol{A}^\top \boldsymbol{A} + \lambda \boldsymbol{I}$ is positive definite, and hence invertible, for every $\lambda > 0$, even when the columns of $\boldsymbol{A}$ are linearly dependent (for example, when $m < n$). Why does this matter in contrast with the least-squares solution $(\boldsymbol{A}^\top \boldsymbol{A})^{-1} \boldsymbol{A}^\top \boldsymbol{b}$?

(d) **[code]** Compute the ridge regression solution for the data set in Problem 2(d) with $\lambda = 0.001$. What is the residual sum of squares $\|\boldsymbol{A}\boldsymbol{x}^\star - \boldsymbol{b}\|^2$?

6. **[code]** The *cross entropy* between a label $y \in \{0, 1\}$ and a predicted probability $p \in (0, 1)$ that the label is 1 is

$$H(y, p) = -y \log p - (1 - y) \log(1 - p).$$

Logistic regression predicts the probability that $y_i = 1$ as $p_i = \sigma(\boldsymbol{a}_i^\top \boldsymbol{x})$, where $\boldsymbol{a}_i \in \mathbb{R}^n$ is the feature vector of data point i, $\boldsymbol{x} \in \mathbb{R}^n$ are the coefficients, and $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid function. The coefficients are computed by minimizing the total cross entropy $L(\boldsymbol{x}) = \sum_{i=1}^m H(y_i, \sigma(\boldsymbol{a}_i^\top \boldsymbol{x}))$ over the m data points.

(a) Show that $H(y_i, p_i)$ is the negative log-likelihood of observing y_i when y_i is a Bernoulli random variable with $\Pr[y_i = 1] = p_i$. Then show that the loss simplifies to

$$L(\boldsymbol{x}) = \sum_{i=1}^m \left(\log\left(1 + e^{\boldsymbol{a}_i^\top \boldsymbol{x}}\right) - y_i \, \boldsymbol{a}_i^\top \boldsymbol{x} \right).$$

(b) Show that $\sigma'(z) = \sigma(z)(1 - \sigma(z))$ and use it to derive the gradient

$$\nabla L(\boldsymbol{x}) = \sum_{i=1}^m \left(\sigma(\boldsymbol{a}_i^\top \boldsymbol{x}) - y_i \right) \boldsymbol{a}_i = \boldsymbol{A}^\top \left(\sigma(\boldsymbol{A}\boldsymbol{x}) - \boldsymbol{y} \right),$$

where the rows of $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ are $\boldsymbol{a}_i^\top$, $\boldsymbol{y} = (y_1, \ldots, y_m)^\top$, and σ is applied to each element of $\boldsymbol{A}\boldsymbol{x}$.

(c) Consider the following eight data points (t_i, y_i):

t_i	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y_i	0	0	0	1	0	1	1	1

Fit the logistic regression model $\Pr[y = 1 \mid t] = \sigma(x_1 + x_2 t)$ with an intercept x_1 and slope x_2. Construct the matrix $\boldsymbol{A}$ and implement gradient descent *from scratch* using the gradient from part (b); do **not** use a logistic regression library (such as `GLM.jl`, `scikit-learn`, or R's `glm`). Report the step size, the number of iterations, and the coefficients $\boldsymbol{x}^\star = (x_1^\star, x_2^\star)^\top$ you obtain, and plot the loss $L(\boldsymbol{x}_k)$ against the iteration number k.

(d) Plot the eight data points together with the predicted probability $\sigma(x_1^\star + x_2^\star t)$ as a function of $t \in [0, 4.5]$. At what value of t does the model predict $\Pr[y = 1 \mid t] = 0.5$?

7. [*Hard*] Consider the quadratic function $f\colon \mathbb{R}^n \to \mathbb{R}$ defined as

$$f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{x},$$

where $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ is symmetric positive definite with eigenvalues $0 < \lambda_{\min} = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_n = \lambda_{\max}$, and let $\boldsymbol{x}^\star = \boldsymbol{A}^{-1}\boldsymbol{b}$ be its unique minimizer. Gradient descent with a constant step size $\alpha > 0$ computes $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \alpha \nabla f(\boldsymbol{x}_k)$ starting from some $\boldsymbol{x}_0 \in \mathbb{R}^n$. Let $\boldsymbol{e}_k = \boldsymbol{x}_k - \boldsymbol{x}^\star$ denote the error at iteration k.

(a) Show that the error satisfies $\boldsymbol{e}_{k+1} = (\boldsymbol{I} - \alpha \boldsymbol{A})\, \boldsymbol{e}_k$ and, therefore, $\boldsymbol{e}_k = (\boldsymbol{I} - \alpha \boldsymbol{A})^k \boldsymbol{e}_0$.

(b) Using the eigendecomposition $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top$, where $\boldsymbol{Q}$ is orthogonal, show that

$$\|\boldsymbol{e}_k\| \le \rho(\alpha)^k \, \|\boldsymbol{e}_0\|, \qquad \text{where} \qquad \rho(\alpha) = \max_{i=1,\dots,n} |1 - \alpha\lambda_i| = \max\left\{|1 - \alpha\lambda_{\min}|,\ |1 - \alpha\lambda_{\max}|\right\}.$$

Hint: Express $\boldsymbol{e}_k$ in the basis of the eigenvectors of $\boldsymbol{A}$ and recall that multiplying by an orthogonal matrix does not change the Euclidean norm.

(c) Show that gradient descent converges to $\boldsymbol{x}^\star$ from every starting point $\boldsymbol{x}_0$ if and only if $0 < \alpha < 2/\lambda_{\max}$. What happens to the iterates when $\alpha > 2/\lambda_{\max}$?

(d) Find the step size $\alpha^\star$ that minimizes $\rho(\alpha)$ and show that

$$\rho(\alpha^\star) = \frac{\kappa - 1}{\kappa + 1}, \qquad \text{where} \qquad \kappa = \frac{\lambda_{\max}}{\lambda_{\min}}$$

is the condition number of $\boldsymbol{A}$. Approximately how many iterations are needed to reduce the error by a factor of 10^{-6} when $\kappa = 10$ and when $\kappa = 1000$? Explain what this implies about gradient descent on poorly conditioned problems.

8. [**Seriously hard**] Consider the same quadratic function f, minimizer $\boldsymbol{x}^\star$, and error $\boldsymbol{e}_k = \boldsymbol{x}_k - \boldsymbol{x}^\star$ as in Problem 7. The *heavy ball method* (Polyak's momentum) adds a multiple of the previous step to the gradient descent update:

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \alpha \nabla f(\boldsymbol{x}_k) + \beta\,(\boldsymbol{x}_k - \boldsymbol{x}_{k-1}),$$

where $\alpha > 0$ is the step size, $\beta \in [0, 1)$ is the momentum parameter, and $\boldsymbol{x}_{-1} = \boldsymbol{x}_0$. In this problem, you will show that a good choice of α and β improves the rate from Problem 7(d) by replacing κ with $\sqrt{\kappa}$.

(a) Show that the error satisfies $\boldsymbol{e}_{k+1} = ((1 + \beta)\boldsymbol{I} - \alpha \boldsymbol{A})\, \boldsymbol{e}_k - \beta \boldsymbol{e}_{k-1}$. Then show that the stacked error $\boldsymbol{z}_k = \begin{bmatrix} \boldsymbol{e}_k \\ \boldsymbol{e}_{k-1} \end{bmatrix} \in \mathbb{R}^{2n}$ satisfies $\boldsymbol{z}_{k+1} = \boldsymbol{T}\boldsymbol{z}_k$ for

$$\boldsymbol{T} = \begin{bmatrix} (1 + \beta)\boldsymbol{I} - \alpha \boldsymbol{A} & -\beta \boldsymbol{I} \\ \boldsymbol{I} & \boldsymbol{0} \end{bmatrix}.$$

(b) Using the eigendecomposition $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top$, show that the eigenvalues of $\boldsymbol{T}$ are exactly the eigenvalues of the n matrices

$$\boldsymbol{T}_i = \begin{bmatrix} 1 + \beta - \alpha\lambda_i & -\beta \\ 1 & 0 \end{bmatrix}, \qquad i = 1, \dots, n,$$

and that the eigenvalues μ of $\boldsymbol{T}_i$ are the roots of $\mu^2 - (1 + \beta - \alpha\lambda_i)\mu + \beta = 0$. *Hint*: Change variables to $\tilde{\boldsymbol{e}}_k = \boldsymbol{Q}^\top \boldsymbol{e}_k$ and observe that the recursion decouples into n independent scalar recursions, one for each coordinate.

(c) Show that if $(1-\sqrt{\beta})^2 \le \alpha\lambda_i \le (1+\sqrt{\beta})^2$, then both roots of the quadratic in part (b) satisfy $|\mu| = \sqrt{\beta}$. *Hint*: Show that the discriminant is non-positive; the roots are then complex conjugates (or a repeated real root), and their product is β.

(d) Let $\kappa = \lambda_{\max}/\lambda_{\min}$ and choose

$$\alpha = \frac{4}{\left(\sqrt{\lambda_{\max}} + \sqrt{\lambda_{\min}}\right)^2}, \qquad \beta = \left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^2.$$

Show that the condition in part (c) holds for every eigenvalue λ_i of $\boldsymbol{A}$, and conclude that every eigenvalue of $\boldsymbol{T}$ has modulus

$$\sqrt{\beta} = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}.$$

You may use without proof the fact that for any square matrix $\boldsymbol{T}$ whose eigenvalues all have modulus at most r and any $\epsilon > 0$, there exists a constant C such that $\|\boldsymbol{T}^k \boldsymbol{z}\| \le C(r+\epsilon)^k \|\boldsymbol{z}\|$ for all k and $\boldsymbol{z}$. Approximately how many iterations does the heavy ball method need to reduce the error by a factor of 10^{-6} when $\kappa = 10$ and when $\kappa = 1000$? Compare with your answer for gradient descent in Problem 7(d).

9. [*Easy, OK to skip if familiar*] ISL **2nd edition**, Chapter 3, Problem 8. You can use a library or your code. Get the dataset from here `https://www.statlearning.com/s/Auto.csv`. Note that missing values of `horsepower` are recorded as `?`; treat them as missing (for example, using `na_values="?"` in pandas or `na.strings = "?"` in R) and remove those rows.

10. [*Easy, OK to skip if familiar*] ISL **2nd edition**, Chapter 4, Problem 14, parts: a, c, f only. The same dataset as above.

11. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).